MediBridge

Hyperlocal Medicine Availability Platform

Find the medicine.
Not another pharmacy.

ARCHITECTURE & TECHNICAL APPROACH

Reforged '26

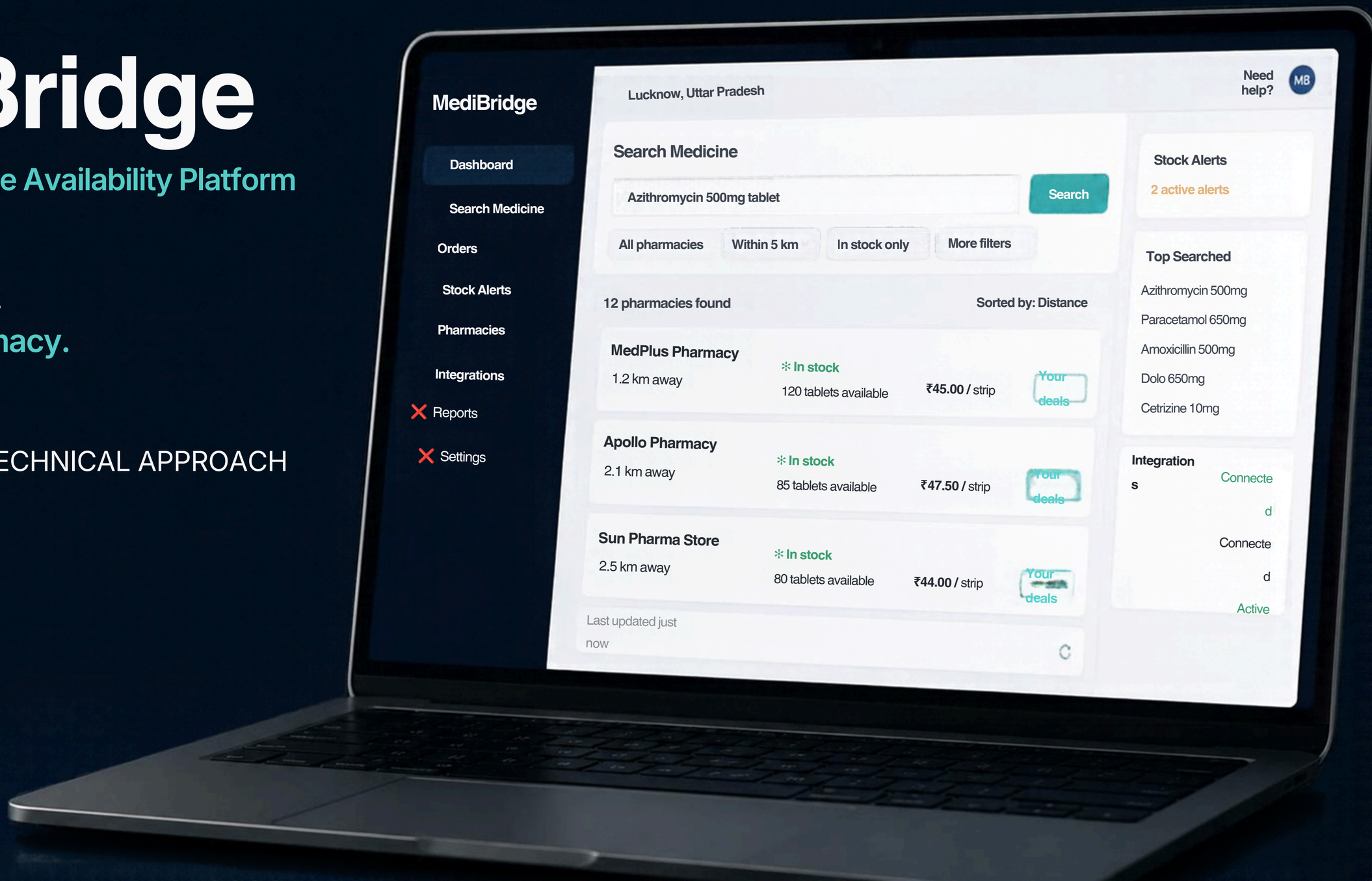
TEAM MEMBERS :-

Shourya Thakur

Shubhansh Upadhyay

Anjali Mishra

Dhairya Tyagi



How MediBridge is built ?

Four connected layers powering hyperlocal medicine availability :-

1. USER INTERFACE

Streamlit dashboard

Patient and chemist interactions

2. APPLICATION LAYER

Python business logic

Request handling and permissions

Workflow coordination

3. AI AND DATA-PROCESSING LAYER

Understanding medicine requests

English/Hinglish message understanding

Medicine intent and information extraction

Structured output generation

Input validation and review flags

4. DATABASE LAYER

Reliable and structured data

SQLite

SQLAlchemy

Medicine, pharmacy, inventory, and stock history records

INTEGRATIONS

TinyFish API

Google Maps links

Scheduling and notification services

How MediBridge understands medicine messages

From natural language input to validated inventory updates :-

1 NATURAL-LANGUAGE INPUT

From chemist

"Paracetamol ke 20 strips available hain."

From patient identification

"Paracetamol 500mg kahan available hai?"

2 LANGUAGE UNDERSTANDING

English/Hinglish medicine-related message parsing

Intent identification

3 INFORMATION EXTRACTION

Extract medicine name, stock action, quantity

Illustrative example:

Input:
"Paracetamol ke 20 strips available hain."

Structured info:
Medicine: Paracetamol
Action: Stock update
Quantity: 20 strips

4 VALIDATION AND BUSINESS RULES

Python logic checks extracted info for processability

Applies business rules and data checks

5 REVIEW DECISION

NEEDS REVIEW?

YES → Flag for human review or confirmation

NO

6 INVENTORY UPDATE

Validated info updates inventory database

Used for patient searches and availability

Message → Language understanding → Structured extraction → Validation → Review decision → Inventory

AI / LANGUAGE PROCESSING
Understands and extracts meaning

PYTHON LOGIC
Applies validation and business rules

DATABASE OPERATIONS
Updates inventory for real-time availability

AI helps interpret the message; validation and review rules control how the extracted information is used.

Illustrative processing workflow — not an actual system log.

From a medicine request to useful results

Two connected workflows delivering reliable, hyperocal availability :-

PATIENT WORKFLOW

Search medicine → Match medicine details → Check inventory → Display nearby pharmacy options

Show reported stock status and latest update time where available

CHEMIST WORKFLOW

Submit stock message → Process message → Validate information → Update inventory → Refresh availability results

SHARED INVENTORY DATABASE

Single source of reported availability and stock updates

RELIABILITY PRINCIPLES

- Track stock freshness
- Flag uncertain updates for review
- Distinguish reported availability from confirmed stock

SIDE INTEGRATIONS

TinyFish API
pharmacy information discovery

Google Maps links
navigation

Scheduling and notifications
background tasks and alerts

Finding a pharmacy online does not prove that a medicine is in stock.

Pharmacist confirmation required for medicine substitutions

Chemist confirmation required for SOS replies

Strengthen the core. Expand responsibly.

01



IMPROVE

Medicine matching

Message validation

02



CONNECT

Standardised stock updates

Repeatable pharmacy onboarding

03



STRENGTHEN

Stock freshness checks

Database performance

Reliability and monitoring

04



EXPAND

Wider pharmacy coverage

Additional integrations

Infrastructure improvements as required

One connected workflow. Clearer medicine availability.